

Vicious Bias

There is a vicious attribute to the bias: it can hide best when its impact is largest. Owing to the invisibility of the dead rats, the more lethal the risks, the less visible they will be, since the severely victimized are likely to be eliminated from the evidence. The more injurious the treatment, the larger the difference between the surviving rats and the rest, and the more fooled you will be about the *strengthening* effect. One of the two following ingredients is necessary for this difference between the true effect (weakening) and the observed one (strengthening): a) a degree of inequality in strength, or diversity, in the base cohort, or b) unevenness, or diversity, somewhere in the treatment. Diversity here has to do with the degree of uncertainty inherent in the process.

More Hidden Applications

We can keep going with this argument; it has such universality that once we get the bug it is hard to look at reality with the same eyes again. Clearly it robs our observations of their realistic power. I will enumerate a few more cases to illustrate the weaknesses of our inferential machinery.

The stability of species. Take the number of species that we now consider extinct. For a long time scientists took the number of such species as that implied from an analysis of the extant fossils. But this number ignores the silent cemetery of species that came and left without leaving traces in the form of fossils; the fossils that we have managed to find correspond to a smaller proportion of all species that came and disappeared. This implies that our biodiversity was far greater than it seemed at first examination. A more worrisome consequence is that the rate of extinction of species may be far greater than we think—close to 99.5 percent of species that transited through earth are now extinct, a number that scientists have kept raising through time. Life is a great deal more fragile than we have allowed for. But this does not mean we (humans) should feel guilty for extinctions around us; nor does it mean that we should act to stop them—species were coming and going before we started messing up the environment. There is no need to feel moral responsibility for every endangered species.

Does crime pay? Newspapers report on the criminals who get caught. There is no section in *The New York Times* recording the stories of those who committed crimes but have not been caught. So it is with cases of tax evasion, government bribes, prostitution rings, poisoning of wealthy spouses (with substances that do not have a name and cannot be detected), and drug trafficking.